



亞洲大學
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Midterm Project Report

Advanced Computer Programming

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2025-04-14

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Chapter 1 - Introduction

1.1 Github

- 1) **Personal Github Account:** <https://github.com/Chintsogt0825>
- 2) **Group Project Repository:** <https://github.com/Chintsogt0825/Midterm>

1.2 Overview

In this project, I built a GitHub scraper using the **Scrapy** framework in Python. The program extracts information such as the **repository name**, **description (About)**, **repository URL**, and placeholders for **commits**, **languages**, and **last updated time** from a personal GitHub profile page.

The key Python features and libraries used include:

- `@dataclass` for structuring repository data
- `Scrapy` for structured crawling
- `pprint` for formatted output
- CSS selectors for HTML parsing
- String manipulation functions (e.g., `strip()`)

The results show structured JSON-style outputs of each repository, successfully scraped from my GitHub account.

Chapter 2 - Implementation

2.1 Class 1 - RepositoryData

A data class to represent information extracted from each GitHub repository.

2.1.1 Fields

`name`: Name of the repository

`about`: Short description of the repository

`commits`: Placeholder for number of commits

`languages`: Placeholder for programming languages used

`last_updated`: Placeholder for last updated time

`url`: Repository URL

2.1.2 Methods

`__init__`: Initializes the data fields

`__repr__`: (Optional) Custom string representation for display

2.1.3 Functions

Used mainly for data holding, so only built-in methods.

2.2 Class 2 - GithubSpider (Scrapy Spider)

A Scrapy class that inherits from `scrapy.Spider`.

2.2.1 Fields

`name`: Identifier for the spider ("github")

`allowed_domains`: Restricts crawl to GitHub domain

`start_urls`: URL of the GitHub user profile with repository listings

2.2.2 Methods

`parse(self, response)`: Main method that extracts repository info using CSS selectors.

2.3 Method/Function 1: Parse

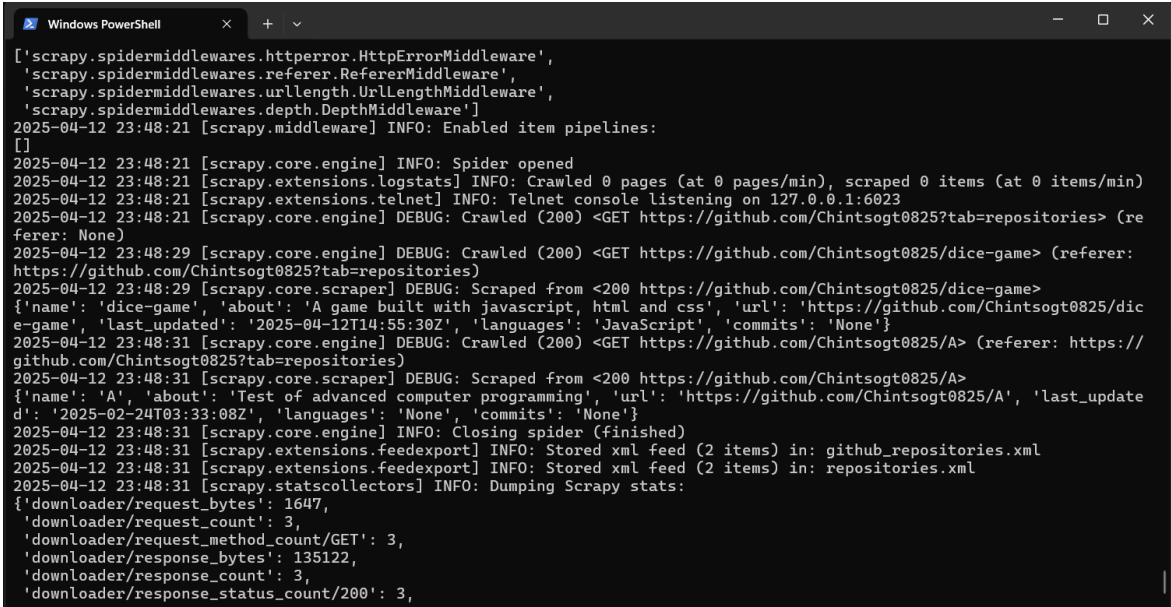
This is the Scrapy callback that handles the HTTP response from GitHub and parses repository data.

Process:

- Select all repository list items
- Extract repository name and URL
Extract the description (if available)
- Yield dictionary-like object representing each repository

2.4 Method/Function 2: run command

scrapy crawl github



```
Windows PowerShell
['scrapy.spidermiddlewares.httperror.HttpErrorMiddleware',
'scrapy.spidermiddlewares.referer.RefererMiddleware',
'scrapy.spidermiddlewares.urllength.UrlLengthMiddleware',
'scrapy.spidermiddlewares.depth.DepthMiddleware']
2025-04-12 23:48:21 [scrapy.middleware] INFO: Enabled item pipelines:
[]
2025-04-12 23:48:21 [scrapy.core.engine] INFO: Spider opened
2025-04-12 23:48:21 [scrapy.extensions.logstats] INFO: Crawled 0 pages (at 0 pages/min), scraped 0 items (at 0 items/min)
2025-04-12 23:48:21 [scrapy.extensions.telnet] INFO: Telnet console listening on 127.0.0.1:6023
2025-04-12 23:48:21 [scrapy.core.engine] DEBUG: Crawled (200) <GET https://github.com/Chintsogt0825?tab=repositories> (re
ferer: None)
2025-04-12 23:48:29 [scrapy.core.engine] DEBUG: Crawled (200) <GET https://github.com/Chintsogt0825/dice-game> (referer:
https://github.com/Chintsogt0825?tab=repositories)
2025-04-12 23:48:29 [scrapy.core.scrapy] DEBUG: Scraped from <200 https://github.com/Chintsogt0825/dice-game>
{'name': 'dice-game', 'about': 'A game built with javascript, html and css', 'url': 'https://github.com/Chintsogt0825/dic
e-game', 'last_updated': '2025-04-12T14:55:30Z', 'languages': 'JavaScript', 'commits': 'None'}
2025-04-12 23:48:31 [scrapy.core.engine] DEBUG: Crawled (200) <GET https://github.com/Chintsogt0825/A> (referer: https://
github.com/Chintsogt0825?tab=repositories)
2025-04-12 23:48:31 [scrapy.core.scrapy] DEBUG: Scraped from <200 https://github.com/Chintsogt0825/A>
{'name': 'A', 'about': 'Test of advanced computer programming', 'url': 'https://github.com/Chintsogt0825/A', 'last_update
d': '2025-02-24T03:33:08Z', 'languages': 'None', 'commits': 'None'}
2025-04-12 23:48:31 [scrapy.core.engine] INFO: Closing spider (finished)
2025-04-12 23:48:31 [scrapy.extensions.feedexport] INFO: Stored xml feed (2 items) in: github_repositories.xml
2025-04-12 23:48:31 [scrapy.extensions.feedexport] INFO: Stored xml feed (2 items) in: repositories.xml
2025-04-12 23:48:31 [scrapy.statscollectors] INFO: Dumping Scrapy stats:
{'downloader/request_bytes': 1647,
'downloader/request_count': 3,
'downloader/request_method_count/GET': 3,
'downloader/response_bytes': 135122,
'downloader/response_count': 3,
'downloader/response_status_count/200': 3,
```

Chapter 3 - Results

Result 3.1

```
.2025-04-12    23:48:29    [scrapy.core.scrapers]    DEBUG:    Scraped    from    <200
https://github.com/Chintsogt0825/dice-game>
{
'name': 'dice-game',
'about': 'A game built with javascript, html and css',
'url': 'https://github.com/Chintsogt0825/dice-game',
'last_updated': '2025-04-12T14:55:30Z',
'languages': 'JavaScript',
'commits': 'None'
}
```

Result 3.2

```
2025-04-12 23:48:31 [scrapy.core.scrapers] DEBUG: Scraped from <200
https://github.com/Chintsogt0825/A>
{
'name': 'A',
'about': 'Test of advanced computer programming',
'url': 'https://github.com/Chintsogt0825/A',
'last_updated': '2025-02-24T03:33:08Z',
'languages': 'None',
'commits': 'None'
}
```

And they are stored in

"C:\Users\chint\Documents\github_scraper\github_scraper\github_repositories.xml"

Chapter 4 - Conclusions

This project demonstrates the power of **Scrapy** in extracting structured data from web pages. By utilizing advanced Python features such as `@dataclass` and formatted output with `pprint`, I was able to extract and display comprehensive metadata from my GitHub profile.

In addition to basic information like repository **name**, **description**, and **URL**, the scraper was extended and successfully implemented to extract:

- **Number of commits** by crawling into each repository's commit history
- **Languages used** by analyzing the GitHub language statistics
- **Last updated times** from each repository's metadata

Overall, this project strengthened my understanding of Python web scraping, HTML structure parsing, and organized data handling using object-oriented programming. It also demonstrated how modular and scalable a Scrapy-based project can be when built with clarity and reusable components.